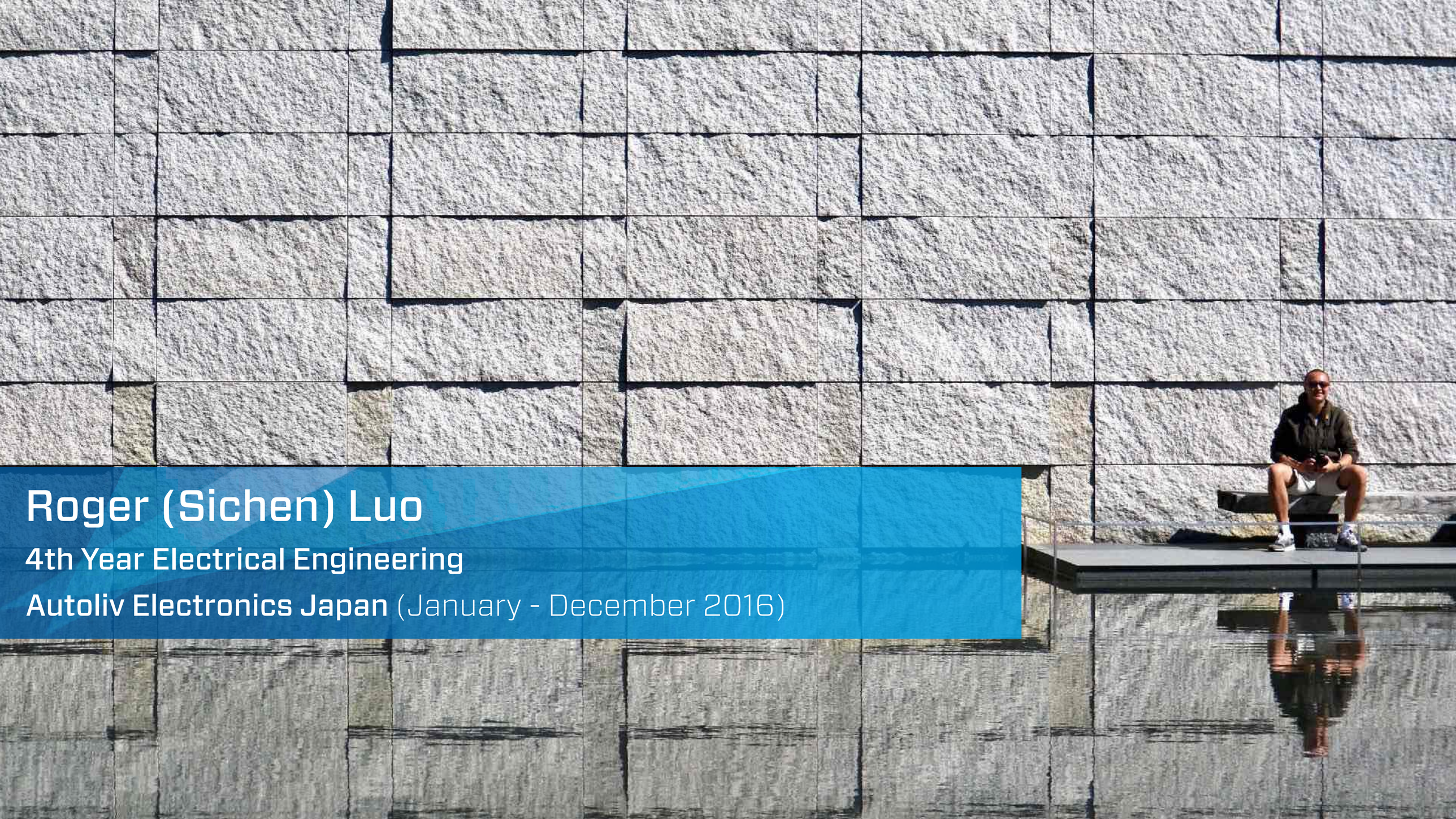


Roger (Sichen) Luo

4th Year Electrical Engineering

Autoliv Electronics Japan (January - December 2016)

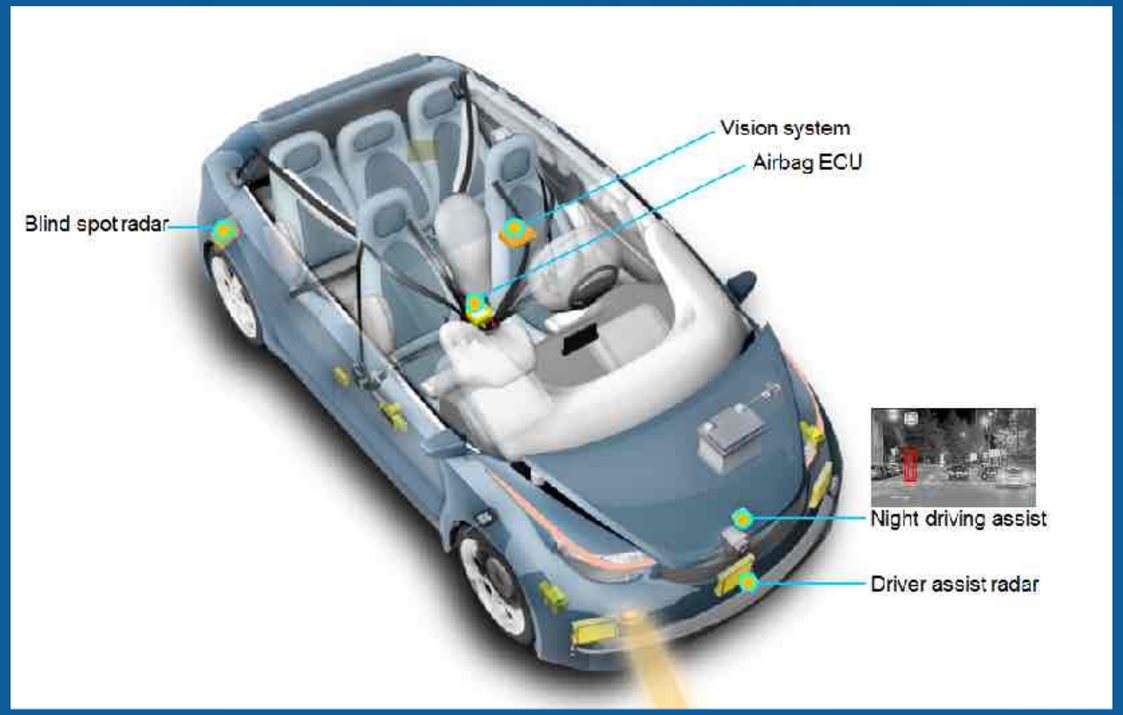
Yokohama
横浜

2nd Largest city in Japan
3M Population
30 min To Tokyo





Organization



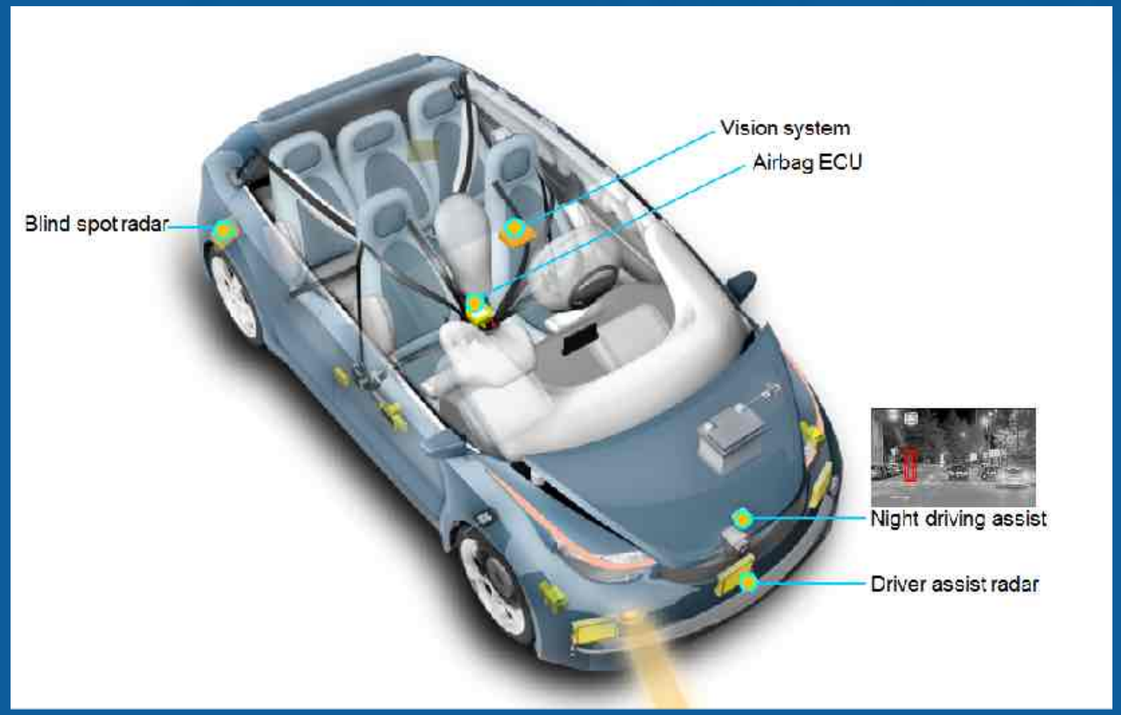
60,000 Employees (Global)

29 Countries

Every year, Autoliv's Products save over 30,000 lives

Autoliv

Organization



100 Employees (Japan)

Passive

Airbag ECU
Airbag Inflator

Active

77GHz Milimeterwave Radar
Mono/Stereo Vision
Driving assistance
Autonomous Driving

Autoliv

Organization



Chelsea and Roger

100 Employees (Japan)

Passive

Airbag ECU
Airbag Inflator

Active

77GHz Milimeterwave Radar
Mono/Stereo Vision

Radar system
Hardware
Software
Features/Functions
Product Assurance

Autoliv

TEST/DEMO VEHICLE SETUP



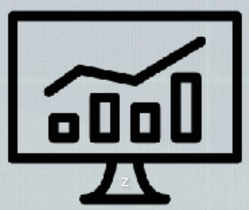
Equipment Rack
Power Distribution
Measurement
Equipment

BENCH TEST



Radar Sensor
Radar Gateway

DATA



Radar Bumper Effect
Radar Tracking
Performance
Radar Cross Traffic Alert
Radar Blind Spot
Detection

DOCUMENTING REPORTING



Data Analysis Criteria
Data Analysis Report
Test Drive checklist
Test Vehicle Specs.
Equipment Prep. Manual

TOOL DEVELOPMENT



Wireless ECU Diagnostics
and CAN signal Monitoring
Automated data acquisition
and File conversion





Data Management and Analysis

Excel Macros
MATLAB
Python

Electronic Hardware

Arduino
Raspberry Pi
Power Distribution
System rack design
Circuit



Computer Software and Programming

Embedded C/C++
Object-oriented Programming
Python
Linux
Real-time

Other Skills

Team Work
Presentation
Critical Thinking
Technical Report

Communication Protocol

CAN CAN-FD FlexRay Automotive Ethernet ISO-TP Time-triggered

Autoliv









Product Assurance Team Diversity

Canadian	Chinese	Filipinos	French	Greek	Indian	Italian	Japanese	Mexican	Swedish
1	1	1	2	1	1	1	8	1	1


2386 E Mall
Vancouver, British Columbia
Google, Inc.
MAY 2016
UBC
UBC ENGINEERING
CO-OP PROGRAM
I want a job!

lots of money

a career

a degree

a job

a house

a car

an iPhone X



lots of money

a career

a degree

a job

a house

a car

an iPhone X



?

A JOURNEY TO DISCOVER YOURSELF

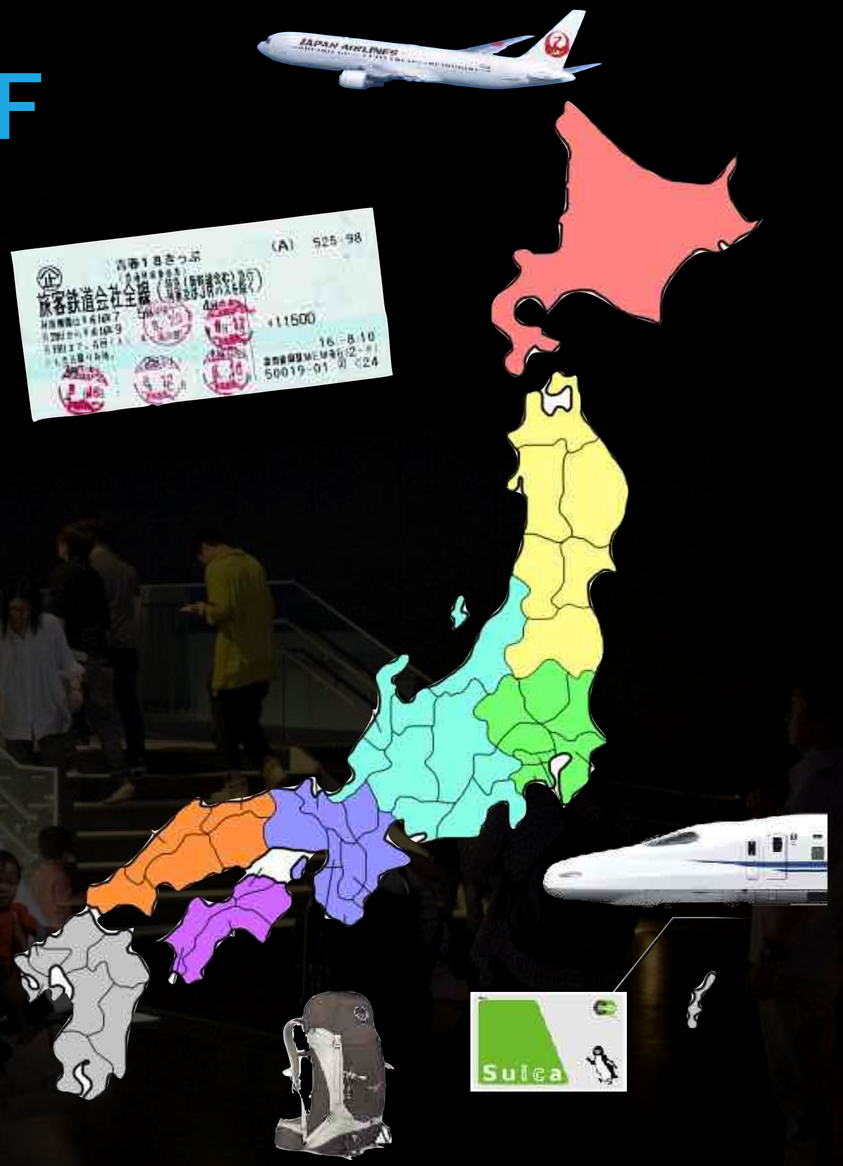
Kanto
Tokyo
Chiba
Ibaraki
Saitama
Kanagawa

Chubu
Gifu
Aichi
Ishikawa
Shizuoka
Kanazawa

Kansai
Kyoto
Nara
Osaka

Chugoku
Hiroshima

Hokkaido
Hakodate
Sapporo
Otaru




Kyoto


奉納


大神






伏見稲荷大社
眷属と授与


國寶
三十三間堂



Nara








adida

Osaka


雪印メグミルク
おいしさと健康
Glico
CHINTAI
LOTTERIA

てっちり
づぼらや本店
ふぐ料理専門


TAKO


Tokyo







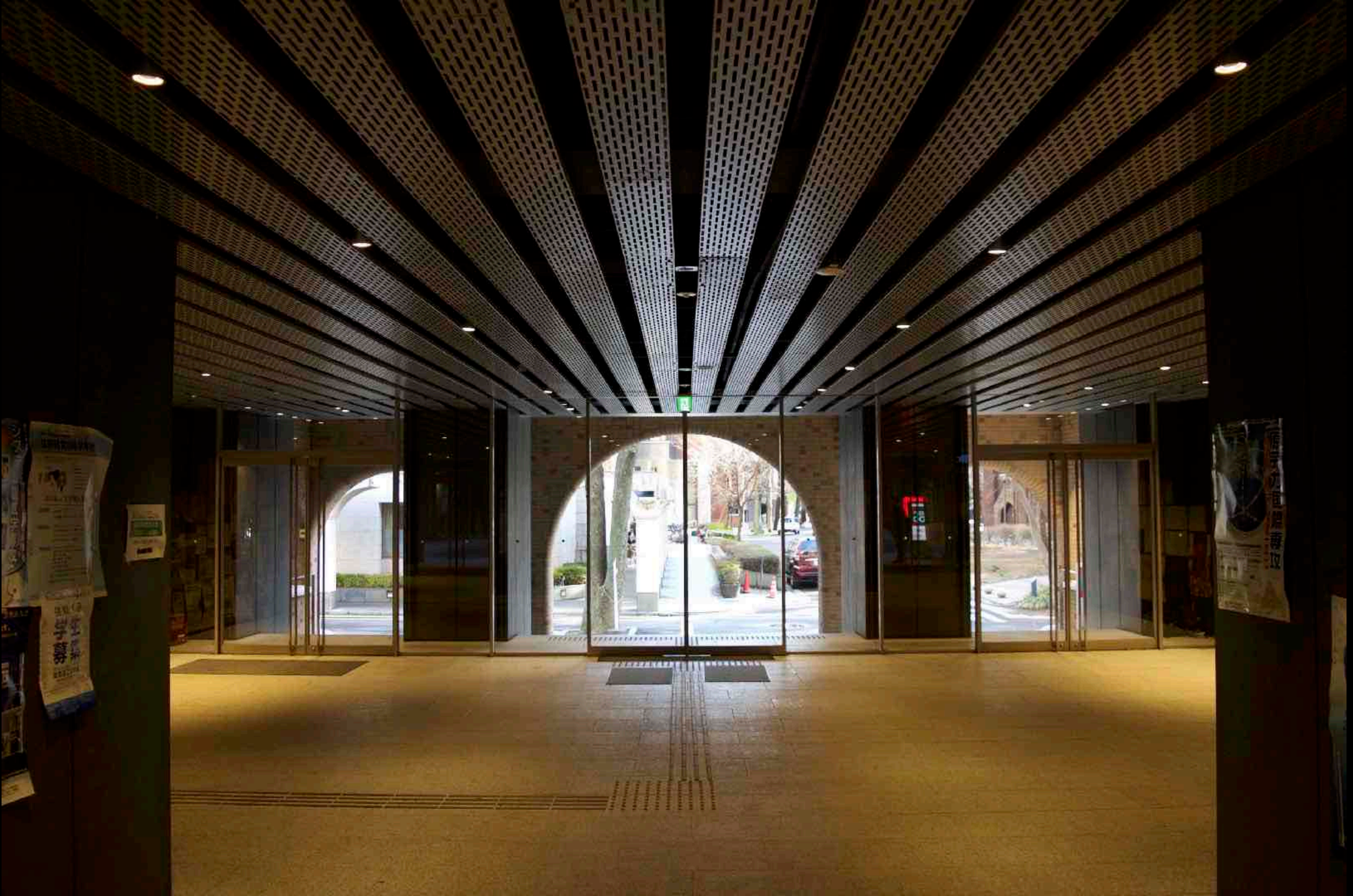
University
of Tokyo



University
of Tokyo



University
of Tokyo

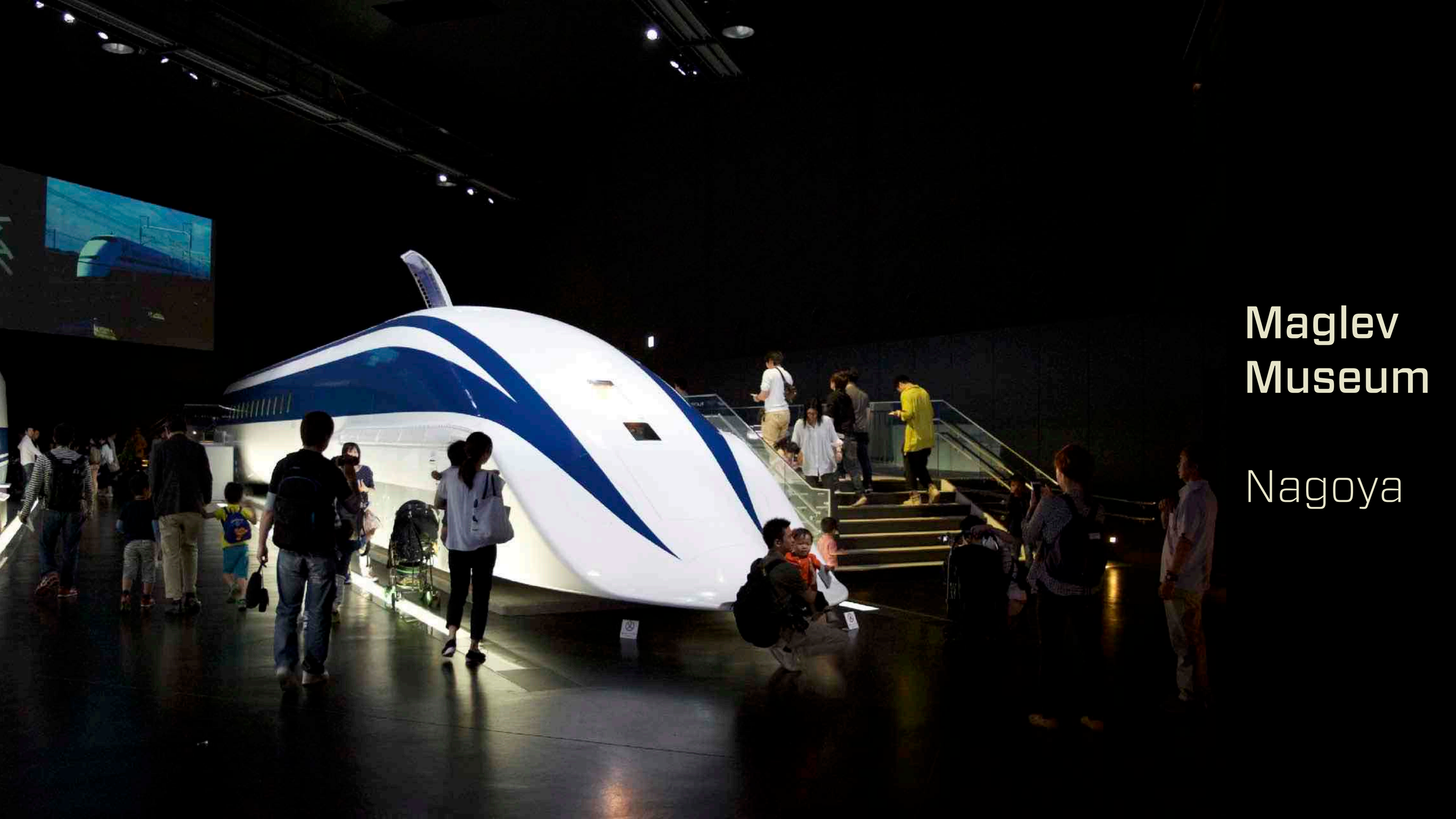


University
of Tokyo

Railway Museum

Tokyo





Maglev Museum

Nagoya

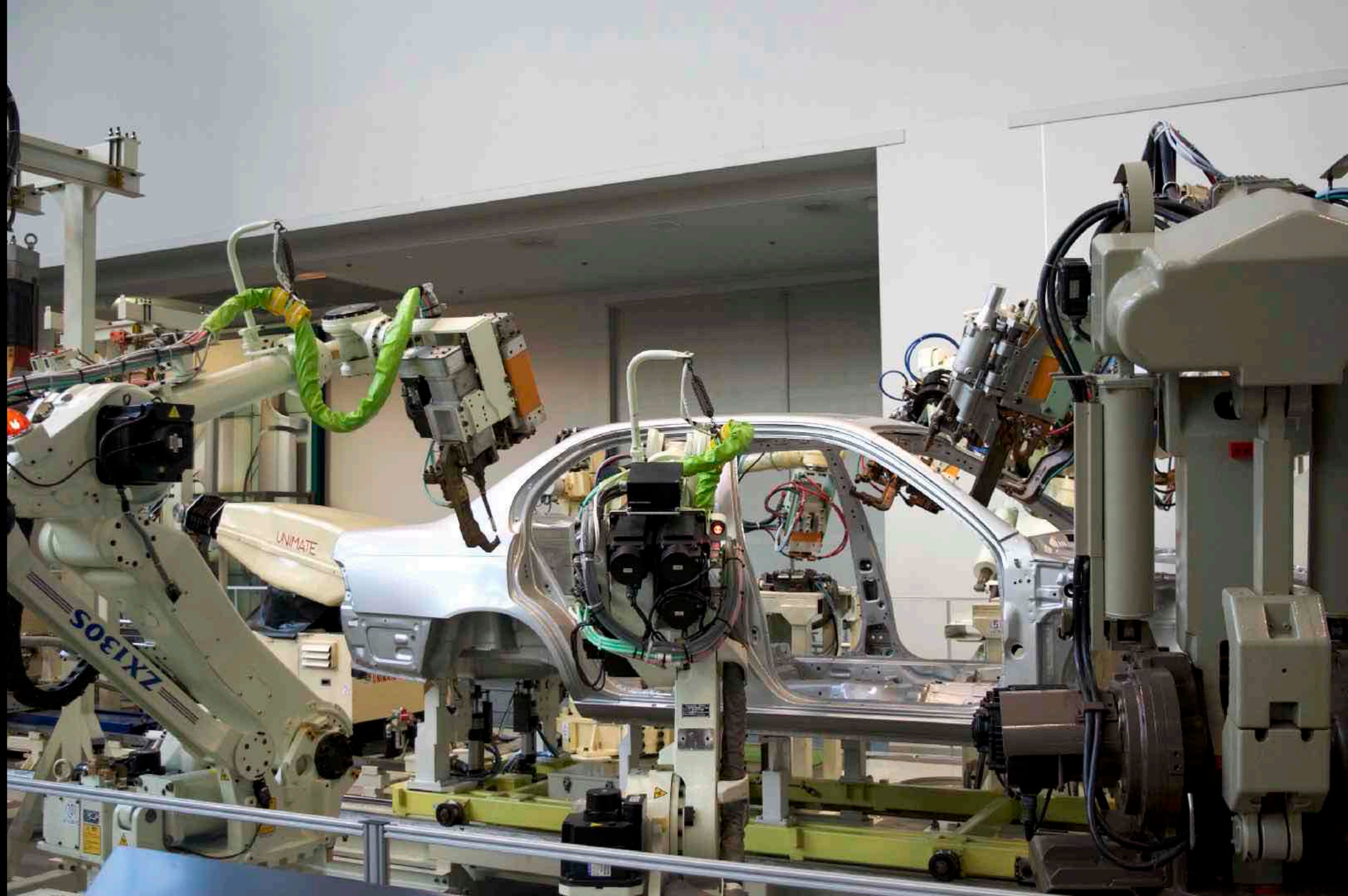


Maglev Museum

Nagoya

Toyota Museum

Nagoya



Toyota Museum

Nagoya





JAL Maintenance Centre

HND Airport

ANA Maintenance Centre

HND Airport




JAL

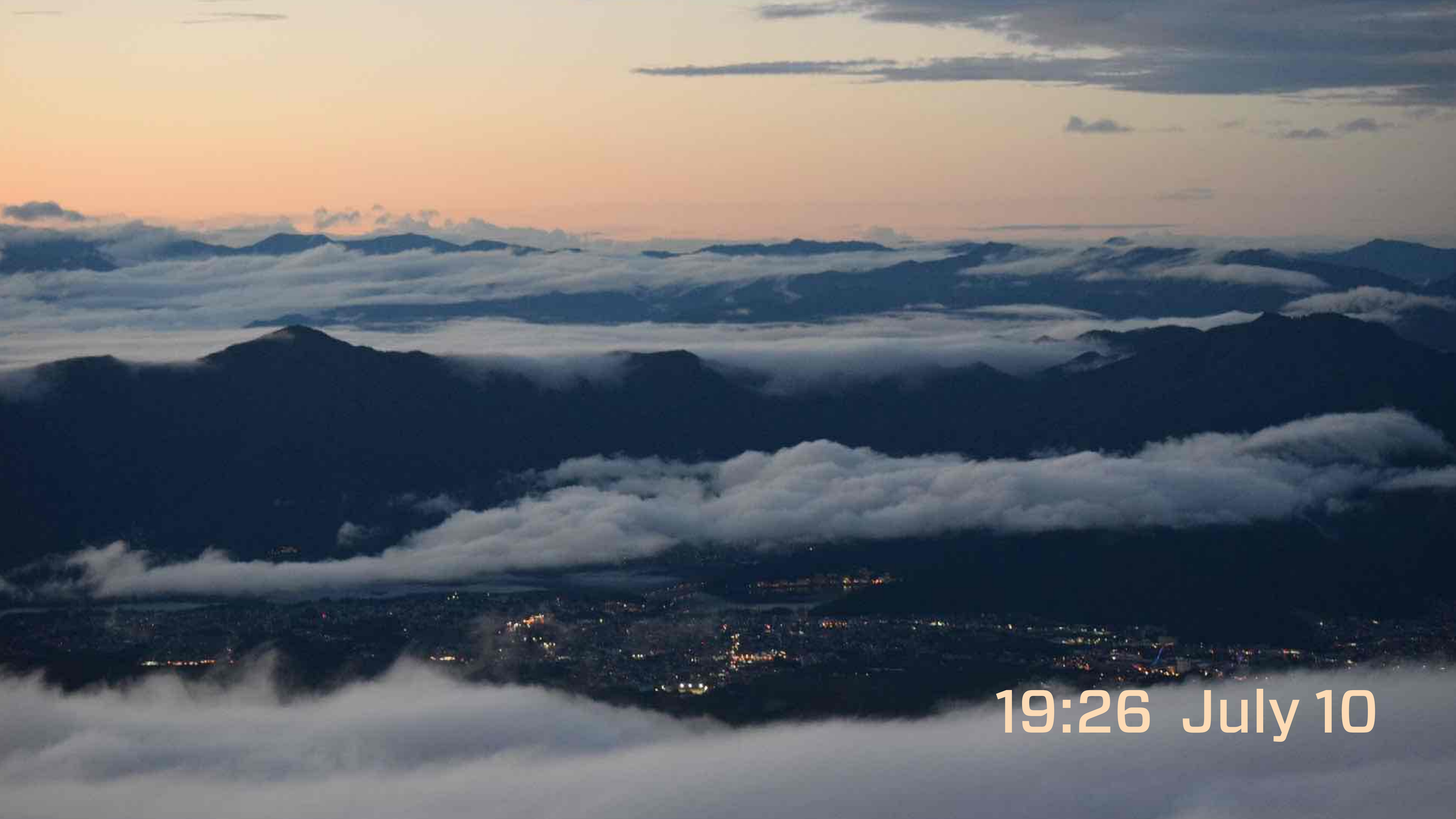


Disney Sea

Fuji-Q Highland




富士山大社 小御嶽神社
2F レストラン・コーヒー
B
19:00 July 10


19:26 July 10


21:00 July 10


3:00 July 11

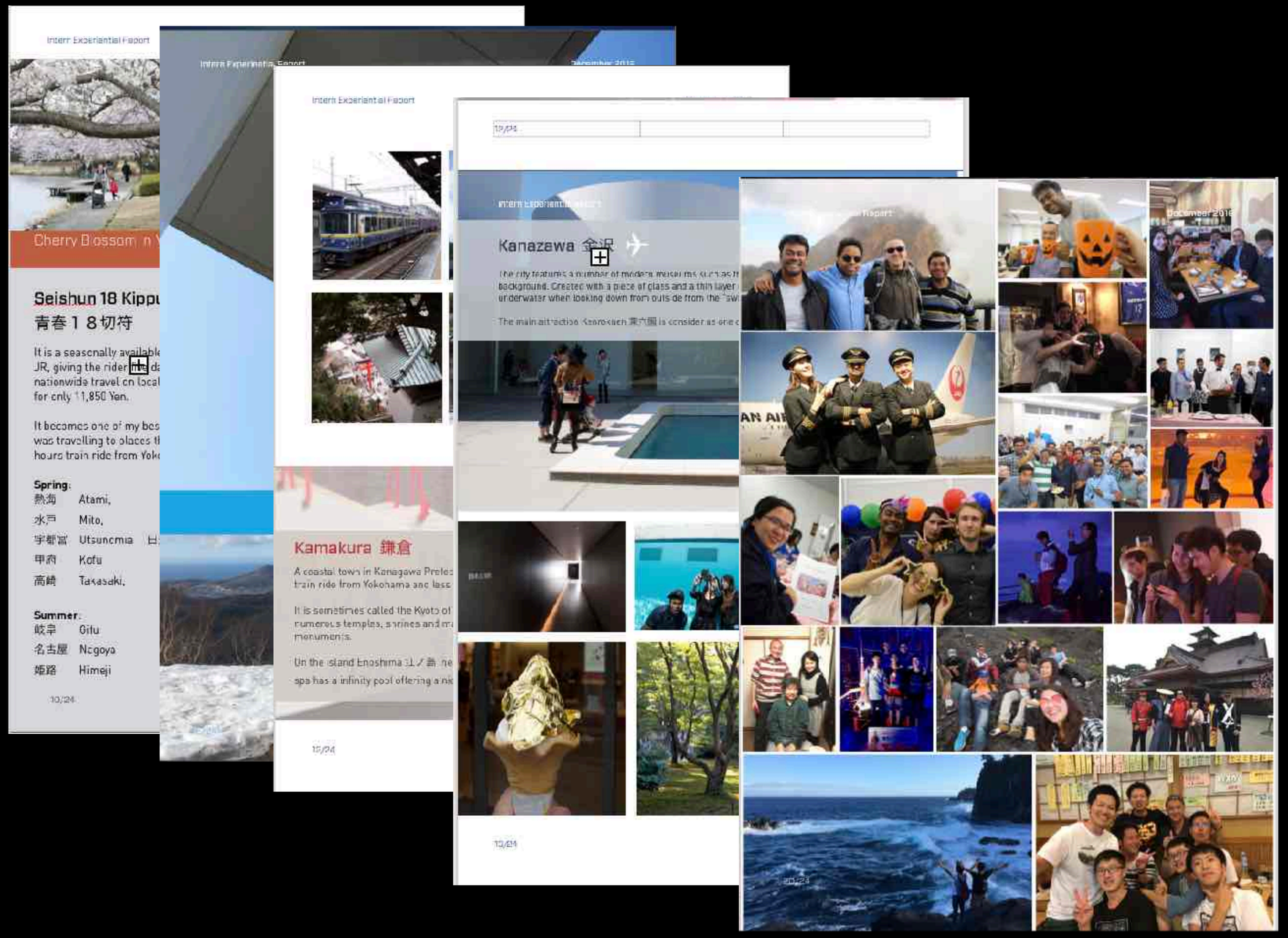

4:30 July 11




富士山頂上浅間大

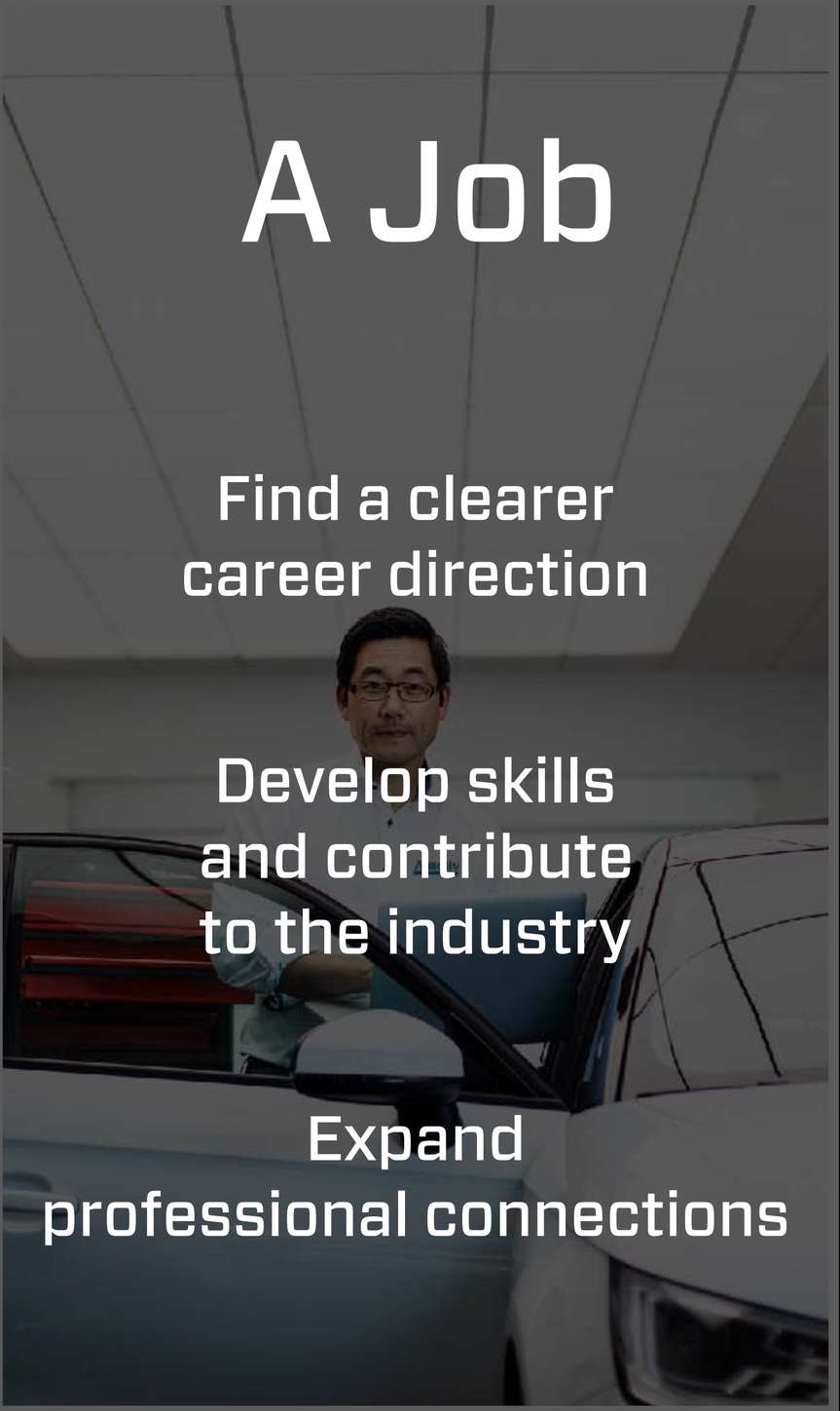





Intern Experiential Report
Cherry Blossom
Seishun 18 Kippu
青春１８切符
It is a seasonally available
JR, giving the rider
nationwide travel on local
for only 11,850 Yen.
It becomes one of my bes
was travelling to places t
hours train ride from Yok
Spring:
熱海 Atami,
水戸 Mito,
宇都宮 Utsunomia
甲府 Kofu
高崎 Takasaki,
Summer:
岐阜 Gifu
名古屋 Nagoya
姫路 Himeji
10/24
Intern Experiential Report
Kamakura 鎌倉
A coastal town in Kanagawa Prefec
train ride from Yokohama and less
It is sometimes called the Kyoto of
numerous temples, shrines and m
monuments.
On the island Enoshima 江ノ島 ne
spa has a infinity pool offering a ne
10/24
Intern Experiential Report
10/24
Kanazawa 金沢
The city features a number of modern museums such as t
background. Created with a piece of glass and a thin layer
underwater when looking down from outside from the "sw
The main attraction Kenrokuen 兼六園 is consider as one
10/24
December 2018

.....

A Job



Find a clearer
career direction

Develop skills
and contribute
to the industry

Expand
professional connections

A Job

Find a clearer career direction

Develop skills and contribute to the industry

Expand professional connections

A Window

Design

Anime

Manga

Craftsmanship

Japanese Quality

Attention to details

Public Transportation

Emerging Technologies

A Job

Find a clearer
career direction

Develop skills
and contribute
to the industry

Expand
professional connections

A Window

Design

Anime

Manga

Craftsmanship

Japanese Quality

Attention to details

Public Transportation

Emerging Technologies

A Platform

Be exposed to
a new language

Be open-minded

Make new friends

Embrace our differences

Appreciate Human’s Creations

Live Different!

lots of money

a career

a degree

a job

a house

a car

an iPhone X



more than a job

lots of money

a career

a degree

a job

a house

a car

an iPhone X



your turn to find out


2386 E Mall
Vancouver, British Columbia
Google, Inc.
MAY 2016
UBC
UBC ENGINEERING
CO-OP PROGRAM
I want a job!



UBC
UBC ENGINEERING CO-OP PROGRAM

APPLY NOW!

THE CANADA-JAPAN CO-OP PROGRAM

FaceBook: Roger Luo
rogerluo1995@gmail.com